

PART 1 HISTORY AND PERSPECTIVE OF INDUSTRIAL ENGINEERING

CHAPTER 1

History of Engineering and Development of Industrial Engineering

1.1. Introduction

Industrial engineering is emerging as one of a small number of “glamor” professions that will be counted on for solving complex problems in the highly technological world of the future. We are seeing rapid advances in the use of automation and other forms of high technology in factories and production systems around the world.

Designing a “factory of the future” is a challenging and complex task—one that requires a knowledge of fundamental sciences, engineering sciences, behavioral sciences, computer and information sciences, economics, and a broad array of topics concerning the basic principles of production systems.

Industrial engineering (I.E.) curricula are designed to prepare students to meet the challenges of the future. Many I.E. graduates will, indeed, design modern manufacturing facilities. Others will elect to design health-care delivery systems, transportation systems, or other systems that provide services rather than manufactured products.

The demand for I.E.s is strong, and growing each year. In fact, the demand for I.E.s greatly exceeds the supply. This demand/supply imbalance is greater for

I.E. than for any other engineering or science discipline and is projected to exist for many years in the future.¹

The purpose of this book is to provide an introduction and overview of the exciting field of industrial engineering. Our intent is to provide a framework, so that the student may have a basic frame of reference as he or she encounters in-depth courses in each of the specialty areas within I.E.

1.2. Early Developments

How did the two words "industrial" and "engineering" become combined to form the label "industrial engineering"? What is the relationship of industrial engineering to other engineering disciplines, to business administration, to the social sciences?

To understand the role of industrial engineering in today's complex world, it is helpful to learn the historical developments that were instrumental in the evolution of I.E. There are many ways to write a history of engineering. The treatment in this chapter is brief because our interest is in reviewing the highlights of engineering development, particularly those leading to industrial engineering as a specialty. More complete histories are available in the references.²⁻⁴

Engineering and science have developed in a parallel, complementary fashion, although not always at the same pace. Whereas science is concerned with the quest for basic knowledge, engineering is concerned with the application of scientific knowledge to the solution of problems and to the quest for a "better life." Obviously, knowledge cannot be applied until it is discovered, and once discovered, it will soon be put to use. In its efforts to solve problems, engineering provides feedback to science in areas where new knowledge is needed. Thus, science and engineering work hand in hand.

Although "science" and "engineering" each have distinguishing characteristics and are regarded as different disciplines, in some cases a "scientist" and an "engineer" might be the same person. This was especially true in earlier times when there were very few means of communicating basic knowledge. The person who discovered the knowledge also put it to use.

A complete chronology of early engineering practice is beyond the scope of this text. The interested reader will find an excellent treatise in *The Ancient Engineers* by de Camp (footnote 2), in which numerous early developments are described and illustrated.

¹Science and Engineering Education for the 1980's and Beyond, prepared by the National Science Foundation and the Department of Education, Oct. 1980.

²L. S. de Camp, *The Ancient Engineers* (Cambridge, Mass.: MIT Press, 1963).

³J. D. Kemper, *The Engineer and His Profession*, 2nd ed. (New York: Holt, Rinehart and Winston, 1975).

⁴R. J. Smith, *Engineering as a Career*, 3rd ed. (New York: McGraw-Hill Book Company, 1969).

We naturally think of such outstanding accomplishments as the pyramids, the Great Wall of China, the Roman construction projects, and so forth, when we think of early engineering accomplishments. Each of these involved an impressive application of fundamental knowledge.

Just as fundamental, however, were accomplishments that are not as well known. The inclined plane, the bow, the wheel, the corkscrew, the waterwheel, the sail, the simple lever, and many, many other developments were very instrumental in the engineer's efforts to provide a better life.

Almost all engineering developments prior to 1800 had to do with physical phenomena: overcoming friction, lifting, storing, hauling, constructing, fastening. Later developments were concerned with chemical and molecular phenomena: electricity, properties of materials, thermal processes, combustion, and other chemical processes.

Fundamental to almost all engineering developments were the advances made in mathematics. Procedures for accurately measuring distances, angles, weights, and time were necessary for almost all early engineering accomplishments. As these procedures were refined, greater accomplishments were realized.

Another very important contribution of mathematics was the ability to represent reality in abstract terms. A mathematical model of a complex system can be manipulated such that relationships between variables in the system can be understood. The simple relationship commonly called the *Pythagorean theorem* is such an example. This theorem states that the hypotenuse of a right triangle is expressed as the square root of the sum of the squares of the adjacent sides. The use of abstract models representing complex physical systems is a fundamental tool of engineers.

As a final comment on early developments, let us discuss an early development that *did not* occur. The missing early development is related to the behavioral sciences. The understanding of human behavior has lagged considerably behind developments in the mathematical, physical, and chemical sciences. This is important to industrial engineers because the systems designed by I.E.s involve people as basic components. The lack of progress in behavioral science has hindered the industrial engineer in his efforts to design optimal systems involving people.

1.3. The Modern Era

We shall arbitrarily define the *modern* era of engineering as beginning in 1750, even though there were many important developments between 1400 and 1750. We choose 1750 as the beginning of modern engineering for two reasons:

- (1) Engineering schools appeared in France in the eighteenth century.
- (2) The term *civil engineer* was first used in 1750.

Principles of early engineering were first taught in military academies and were concerned primarily with road and bridge construction and with fortifications. This portion of academic training was referred to as *military engineering*. When

some of the same principles were applied to nonmilitary endeavors, it was only natural to refer to these as *civilian engineering*, or simply *civil engineering*.

Interrelated advancements in the fields of physics and mathematics laid the groundwork for practical applications of mechanical principles. An important advancement was the development of a practical steam engine that could perform useful work. Once such an engine was available (approximately 1700), many mechanical devices were developed that could be driven by the engine. These efforts culminated in the emergence of *mechanical engineering* as a distinct branch in the early nineteenth century.

Another example of such an advancement was the fundamental work conducted in the latter part of the eighteenth century on electricity and magnetism. Although early scientists had known about magnetism and static electricity, an understanding of these phenomena did not commence until Benjamin Franklin's famous kite-flying experiment in 1752. The next half-century saw the laying of the foundation of electrical science, primarily by German and French scientists.

The first significant application of electrical science was the development of the telegraph by Samuel Morse (approximately 1840). Thomas Edison's invention of the carbon-filament lamp (approximately 1880) led to widespread use of electricity for lighting purposes. This, in turn, spurred very rapid developments in the generation, transmission, and utilization of electrical energy for a variety of labor-saving purposes. Engineers who chose to specialize in this activity were naturally labeled *electrical engineers*.

Along with the developments in mechanical and electrical technology were accompanying developments in the understanding of substances and their properties. The science of chemistry is concerned with understanding the nature of *matter* and in learning how to produce desirable changes in materials. Fuels were needed for the new internal combustion engines being developed. Lubricants were needed for the rapidly growing array of mechanical devices. Protective coatings were needed for houses, metal products, ships, and so forth. Dyes were needed in the manufacture of a wide variety of consumer products. Somewhat later, synthetic materials were needed to perform certain functions that could not be performed as well or at all by natural materials. This field of engineering endeavor naturally became known as *chemical engineering*.

As industrial organizations emerged to capitalize on the rapidly developing array of technological innovations, the size and complexity of manufacturing units increased dramatically. *Mass production* was made possible through two important concepts:

- Interchangeability of parts.
- Specialization of labor.

Through mass production the unit cost of consumer products was reduced dramatically.

The groundwork was now laid for a dramatic shift in the lifestyles and cultures of industrialized nations. Within half a century the United States and other devel-

oping countries changed from largely rural, agricultural economies and societies to urban, industrialized economies and societies. The suddenness of this change is probably the cause of many of today's pressing problems, for example, pollution and crowding.

During the early part of this movement it was recognized that business and management practices that had worked well for small shops and farms simply were inadequate for large, complex manufacturing organizations. The need for better management systems led to the development of what is now called *industrial engineering*. This development will be traced more completely in later sections of this chapter.

These five major engineering disciplines (civil, chemical, electrical, industrial, and mechanical) were the branches of engineering that emerged prior to the time of World War I. These developments were part of the industrial revolution that was occurring worldwide, and the beginning of the *technological revolution* that is still occurring.

Developments following World War II led to other engineering disciplines, such as *nuclear engineering*, *electronic engineering*, *aeronautical engineering*, and even *computer engineering*. The Space Age led to *astronautical engineering*. Recent concerns with the environment have led to *environmental engineering* and *bioengineering*. These newer engineering disciplines are often considered specialties within one or more of the "big five" disciplines of civil, chemical, electrical, industrial, and mechanical engineering.

1.4. The Engineering Process

We have sketched the development of several engineering branches, and we all have a general understanding about what particular engineers actually do. For example, we know that mechanical engineers are involved with developing better automotive engines and that electrical engineers are concerned with developing better television sets.

Are there characteristics that distinguish engineers from other professionals? If so, what are these characteristics? Let us explore some of the general activities in which all engineers are involved.

- Engineers *solve problems*, but so do mathematicians.
- Engineers *analyze*, but so do statisticians and economists.
- Engineers *design systems*. Do others?

Certainly, architects can legitimately claim that they design systems, if one can call a building a system. To do a complete job of building design requires a knowledge of materials and forces. Thus, it can be argued that only an *architectural engineer* is truly qualified to design a building. Architects who do not have a basic engineering background must rely on engineers for much of their design work.

Within the spirit of the framework being developed, therefore, we can say

that the distinguishing characteristic of engineering is that it is concerned with the design of systems.

What is design? Certainly, design involves a considerable amount of creativity. Is design then simply an art form that one learns through experience? Or are there design principles that can be learned and applied?

Synthesis is a term whose meaning is almost the same as *design*. Perhaps we can develop an understanding of design by discussing the two terms synthesis and analysis.

Analysis is concerned with resolving something into its basic elements; synthesis is concerned with combining elements into a whole. When freshmen engineering students take statics, they learn much about analysis. For example, they are given a beam having certain properties (size, material, loading points, and so forth) and they are asked to analyze the beam in terms of its loading characteristics. The analysis may show, for example, that the beam will support a maximum of 100 tons. There is only one answer to problems involving analysis.

Now, let us consider a different question: Suppose you are asked to design a beam that will support 100 tons. There are a large variety of beams that will support 100 tons; thus, there is not necessarily one best answer to the design problem.

Typically, analysis is concerned with existing systems. Synthesis is usually concerned with a new or improved system.

The engineering process involves both synthesis and analysis. Figure 1.1 is an illustration of the engineering process. Note that this process is used by all engineers: civil, chemical, electrical, industrial, mechanical, and others. The manner in which industrial engineers engage in this process will be discussed in a later section of this chapter.

The dual arrows in Figure 1.1 between the two blocks representing analysis and synthesis are intended to portray the iterative nature of the engineering process. One stage of analysis provides sufficient insight to postulate a particular design. This design, in turn, provides additional information that changes the basis of the analysis. This iterative process is repeated until an acceptable solution, system, or method is derived. (The word "acceptable" is used instead of the word "optimal" because a true optimal design is rarely achieved, and then only within the limitations of the assumptions under which the engineer is working.)

1.5. Engineering as a Profession

Although there can never be general agreement on this matter, the vocations that are commonly referred to as *professions* are medicine, teaching, architecture, law, ministry, and engineering. Smith⁵ points out that these vocations have four common characteristics:

⁵Smith, p. 10.

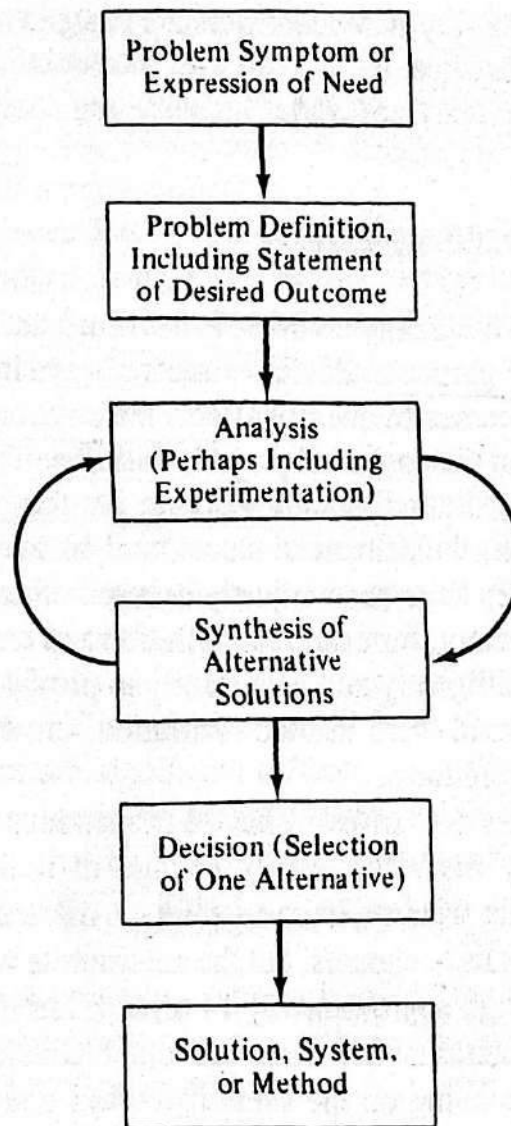


Fig. 1.1. Basic engineering process.

- (1) Associated with a profession is a significant body of special knowledge.
- (2) Preparation for a profession includes an internship-like training period following the formal education.
- (3) The standards of a profession, including a code of ethics, are maintained at a high level through a self-policing system of controls over those practicing the profession.
- (4) Each member of a profession recognizes his responsibilities to society over and above responsibilities to his client or to other members of the profession.

Other definitions of a profession go further and state that its members must engage in *continued study* and that its prime purpose must be rendering of a *public service*. Although these matters are subject to interpretation, they are evidence that professionalism is indeed a serious matter and that much is expected of people who claim to be professionals.

Chapter 4 of Kemper's book (footnote 3) is entitled "Is Engineering Really a Profession?" and is highly recommended reading. Kemper concludes that engineering is indeed a profession. He cautions, however, that "... just because engineering is a profession it does not necessarily follow that every engineer is a

professional. Every individual remains charged with the responsibility to meet the criteria on a personal basis." Kemper also points out that the definition of a professional in the Taft-Hartley Act includes engineers.

* 1.6. Professional Ethics

Engineers are frequently involved in decisions that have a profound effect on society. The design of particular devices almost always involves the safety of the user. The design of processes frequently affects the environment. The design and location of a factory affect the community and its citizens. The design of a management system greatly affects the individuals working for the organization—their comfort, their sense of worth, their financial status, and so on.

Engineers have been unjustly accused of creating almost all the current problems in our society, for example, pollution and crowding. The truth is that engineers have worked diligently and persistently to provide society with the things the social scientists claimed were needed. Pollution, crowding, and unsightly freeways are simply the byproducts.

Engineers do, however, have a tremendous responsibility to protect the public welfare. They must continually engage in trade-offs between costs and factors affecting public welfare, such as safety. An automobile can be made almost completely safe for its occupants, but the automobile would cost approximately \$100,000 and would weigh approximately 14 tons. Even if someone were willing to pay this much for an automobile, there are other considerations: the safety of people in lighter cars traveling on the same roadways and the fuel consumption of a 14-ton vehicle.

Engineers are also frequently caught in a controversy between their employer or client and the public. Companies must produce items for sale at competitive prices, even though a "cheap" design may mean an unsafe or unreliable product.

The engineering profession enjoys a very favorable reputation regarding its adherence to professional ethics. Fortunately, early engineers were perceptive and conscious of the public good. The National Society of Professional Engineers (see Section 1.8) publishes a set of Canons of Ethics for engineers. They do not answer every question or resolve every controversy, but they do provide a good foundation from which one may extrapolate to cover almost any situation. A copy of the Canon of Ethics may be obtained by writing to your State Engineering Registration Board.

* 1.7. Professional Licensing

How would you like to be treated by a medical doctor who had not been examined and approved by your state's licensing board? How would you like to be represented in a court case by a lawyer who had not proven his knowledge of the law by passing the bar exam?

If we agree that doctors and lawyers should be required to be examined before being admitted to practice, let us examine the same general question about the engineering profession. Should bridges be designed by people who have not proven their knowledge of civil engineering? Should airplanes be designed by people who have not proven their knowledge of aerodynamics?

Professional licensing of engineers has not been as rigidly pursued as with the professions of law and medicine. Indeed, many states require barbers to be licensed, but not engineers! One of the reasons for this is that almost all engineers work for large companies, but doctors and lawyers (and even barbers) are usually self-employed.

All states do have provisions for licensing engineers, but this is not required as a condition for practicing engineering, except in certain situations, such as consulting firms. But as companies find themselves more subject to liability suits over product safety and environmental violations, it is expected that almost all engineers will eventually be subject to professional licensing.

Licensing of engineers is typically referred to as *professional registration*. Whereas all 50 states, the District of Columbia, and 4 U.S. territories now have engineering registration laws, there are significant differences in the requirements for professional registration. An engineer can become registered in some states by simply applying and demonstrating that he has performed engineering work for a certain length of time. Other states require extensive examinations on fundamental engineering principles.

Some uniformity appears to be emerging, although complete uniformity may never be realized. The emerging pattern seems to be as follows: Engineering students take the Engineer in Training (E.I.T.) exam during their last year in college; the students graduate, take jobs and practice the engineering profession for a specified period of time; they then take an examination as a final step in the registration process.

A current major issue concerns continuing certification of an engineer's ability. Some states are beginning to consider requiring engineers, doctors, and lawyers to renew their licenses periodically. There is considerable debate whether license renewal should involve reexamination on current engineering material or should simply certify that a certain minimum amount of continuing education and professional development has been achieved.

In view of the current state of uncertainty regarding professional registration, a natural question is "Why should I bother?" Kemper⁶ states six compelling reasons why every engineer should become registered as quickly as possible after graduation:

- (1) Registration will most likely become more important in the future, and will probably be required in some states.
- (2) No young engineer can foresee the total future course of his career: he may believe that he will always be working in areas not requiring registration, but

⁶Kemper, pp. 245-246.

many unexpected events may occur which would require registration in order to practice the engineering profession.

- (3) A court of law generally will not recognize an individual as an engineer unless he is registered, and therefore will not accept an unregistered engineer's testimony as an expert witness.
- (4) More and more states require that an engineer be registered in order to engage in a growing array of activities. If an unregistered person engages in practice required by law to be performed by registered engineers, that person may be penalized and, at the very least, the courts will not aid him in collecting his fee. Many states now prohibit the word "engineering" from appearing in the name of a firm or in its advertisement if it has no registered engineers in responsible positions.
- (5) An increasing number of companies believe it is desirable for members of their engineering management group to be registered; therefore, registration could be an aid in promotion.
- (6) As the passage of written examinations becomes a more universal requirement, registration will likely be increasingly regarded by employers as an indicator of technical competence.

1.8. Engineering Education and ABET Accreditation

The complete education of an engineer occurs in at least three definable stages:

- (1) *Preparatory*—High school courses in mathematics and science.
- (2) *University*—A formal engineering curriculum.
- (3) *Continuing*—Lifelong learning through professional practice and professional development.

Students who are college sophomores in 1993 began their preparatory work in 1988 and will not reach their full growth as practicing engineers until after the year 2000. Engineering education, in its broadest sense, is indeed a long process.

Engineering education has progressed through several stages of evolution. Prior to World War II engineering education was concerned with the art and practice of engineering principles. Engineering students spent long hours learning to operate lathes, drill presses, molding machines, foundries, and so on. They learned to wind motors and to build radio sets. There was a considerable amount of "hands-on" experience involved in the educational process.

The so-called "Grinter Report"⁷ in 1955 instigated significant changes in engineering education. Much of the "art" of engineering was removed from the engineering curricula and was replaced by a much greater emphasis on the basic science underlying engineering. The great achievements in space, communications,

⁷"Report on Evaluation of Engineering Education," American Society for Engineering Education, June 15, 1955.

and other sophisticated systems can be attributed at least in part to the new emphasis on the science content of engineering curricula.

Critics of the aforementioned trend claim that in many cases engineering curricula became almost indistinguishable from curricula in mathematics and physics. The greater emphasis on analysis resulted in a diminished emphasis on engineering design. Consequently, current trends are toward a more balanced approach to engineering education.

Strong cases have been made⁸ for extending the length of engineering curricula to include a year of graduate study leading to the master's degree. Entry into almost all professions (law, medicine, teaching) requires completion of a significant amount of post-baccalaureate education. Proponents of expanded engineering curricula argue that five years of formal training is a minimum time in which to acquire the tools needed for practicing engineering.

Traditionally, graduate education in engineering has led to a "master of science" degree. The emphasis on science education often includes a research effort and a formal thesis. Almost all engineers who receive a master's degree, however, do not go into research or teaching. It appears that there is a need for a path at the master's level that emphasizes *engineering practice* rather than research. Such an approach is often called a *professional engineering program*⁹ and includes a period of professional practice in the real world prior to completion of degree requirements.

The Accreditation Board for Engineering and Technology (ABET) is the official agency in the United States for examining and accrediting engineering curricula. The purpose of ABET accreditation is to assure the public and employers of engineering graduates that certain minimum standards have been met.

Students often ask, "Why should I have to take English, differential equations, or thermodynamics?" Many engineering students are particularly negative toward courses in humanities and the social sciences.

Indeed, why *do* engineering students have to take the particular courses specified on their curriculum sheets? Several reasons can be stated:

- There should certainly be some degree of commonality among the many branches that are called "engineering."
- Companies hiring engineering graduates need some assurance that certain minimum content has been provided. This is especially needed when companies are hiring engineering graduates from several universities in different parts of the country.
- Much of the content of an engineering curriculum is intended as a general foundation from which an engineer may pursue a particular career path. Few engineering students know specifically what they will be doing in the future.
- Courses in humanities and social sciences are included to provide the engineer with an awareness of social values and concerns of nontechnical people in our

⁸M. R. Lohmann, "Professional Schools for Engineers," *Engineering Education*, 60 (1970), p. 954.

⁹Lohmann, p. 954.

society. Also, problems are not just technical in nature; there are always social, economic, and perhaps political factors that must be considered.

The ABET requirements for the curriculum content of a four-year engineering program are as follows:

- The equivalent of approximately two and one-half years of study in the area of mathematics, science, and engineering. The course work should include at least one year of an appropriate combination of mathematics and basic sciences, one year of engineering sciences, and one-half year of engineering design.
- The equivalent of one-half year as the minimum content in the area of the humanities and social sciences.

For those institutions which elect to prepare graduates for entry into the profession at the advanced level, ABET normally requires that students' curricular work (1) satisfy ABET engineering criteria at the basic level for the program being evaluated, and (2) have the equivalent of one additional year of study beyond that required for a basic level program. This additional year of study must include at least two-thirds year of advanced mathematics, basic sciences, engineering science and engineering design. The additional year of study must include a considerable amount of material and treatment at an advanced level not normally associated with the basic level.

It may appear that the ABET is attempting to force a high degree of standardization with the above described criteria. Actually, the ABET encourages experimentation and custom-tailored programs to fit local conditions. The criteria provide considerable flexibility while assuring the inclusion of critical essential elements.

as prod. → productivity → Gantt chart (planning & scheduling of activities)
 → Statistical quality control → inventory → fact design
 * 1.9. Chronology of Industrial Engineering → OR

The preceding sections are intended to portray the broad context in which the field of industrial engineering exists. Although there is considerable commonality among the different branches of engineering, each branch has distinguishing characteristics that are important to recognize. The remainder of this text is concerned with industrial engineering.

Industrial engineering emerged as a profession as a result of the industrial revolution and the accompanying need for technically trained people who could plan, organize, and direct the operations of large complex systems. The need to increase efficiency and effectiveness of operations was also an original stimulus for the emergence of industrial engineering. Some early developments are explored, in order to understand the general setting in which industrial engineering was born. For more details, see the excellent work of Emerson and Naehring.¹⁰

¹⁰Howard P. Emerson and Douglas C. E. Naehring, *Origins of Industrial Engineering* (Atlanta: IE&M Press, 1988).

Charles Babbage visited factories in England and the United States in the early 1800's and began a systematic record of the details involved in many factory operations. For example, he observed that the manufacture of straight pins involved seven distinct operations. He carefully measured the cost of performing each operation as well as the time per operation required to manufacture a pound of pins. Babbage presented this information in a table, and thus demonstrated that money could be saved by using women and children to perform the lower-skilled operations. The higher-skilled, higher-paid men need only perform those operations requiring the higher skill levels. Babbage published a book containing his findings, entitled *On The Economy of Machinery and Manufactures* (1832). In addition to Babbage's concept of division of labor, the book contained new ideas on organizing and very advanced (for the time) concepts of harmonious labor relations. Significantly, Babbage restricted his work to that of observing and did not attempt to improve the work methods or to reduce the operation times.

The concept of *interchangeable manufacture* was a key development leading to the modern system of mass production. This concept was to produce parts so accurately that a specific part of a particular unit of a product could be interchanged with the same part from another unit of the product, with no degradation of performance in either unit of the product. Eli Whitney received a government contract to manufacture muskets using this method. Another of his contributions was the design and construction of new machines that could be operated by laborers having only a minimal amount of training. Through the successful application of these two concepts, Whitney created the first mass production system.

In the period around 1880 industrial operations were conducted in a much different manner from today. There was very little planning and organizing, as such. A first-line supervisor was given verbal instructions on the work to be done and a crew of (usually) poorly trained workers. The supervisor was expected to work his men as hard as he could. Any improved efficiency in work methods usually came from the worker himself in his effort to find an easier way to get his work done. There was virtually no attention given to overall coordination of a factory or process.

Frederick W. Taylor is credited with recognizing the potential improvements to be gained from analyzing the work content of a job and designing the job for maximum efficiency. Taylor's original contribution, constituting the beginning of industrial engineering, was his three-phase method of improving efficiency: Analyze and improve the method of performing work, reduce the times required, and set standards for what the times should be. Taylor's methods brought about significant and rapid increases in productivity. Later developments stemming from Taylor's work led to improvements in the overall planning and scheduling of an entire production process.

Frank B. Gilbreth extended Taylor's work considerably. Gilbreth's primary contribution was the identification, analysis, and measurement of fundamental motions involved in performing work. By classifying motions as "reach," "grasp," "transport," and so on, and by using motion pictures of workers performing their tasks, Gilbreth was able to measure the average time to perform each basic motion

under varying conditions. This permitted, for the first time, jobs to be *designed* and the time required to perform the job known before the fact. This was a fundamental step in the development of industrial engineering as a profession based on "science" rather than "art."

Dr. Lillian Gilbreth, wife of Frank, is credited with bringing to the engineering profession a concern for human welfare and human relations. Having received a doctoral degree in psychology from Brown University, Dr. Gilbreth became a full partner with her husband in developing the foundational concepts of industrial engineering. During Dr. Gilbreth's long life (1878–1972), she witnessed and contributed to the birth, growth, and maturation of the I.E. profession. She became known as the "first lady of engineering" and the "first ambassador of management." She received many, many honors and awards from professional organizations, universities, and governments from around the world. She was the first woman to be elected to the National Academy of Engineering.

Another early pioneer in industrial engineering was Henry L. Gantt, who devised the so-called *Gantt chart*. The Gantt chart was a significant contribution in that it provided a systematic graphical procedure for preplanning and scheduling work activities, reviewing progress, and updating the schedule. Gantt charts are still in widespread use today.

W. A. Shewhart developed the fundamental principles of statistical quality control in 1924. This was another important development in providing a scientific base to industrial engineering practice.

Many other industrial engineering pioneers contributed to the early development of the profession. During the 1920s and 1930s much fundamental work was done on economic aspects of managerial decisions, inventory problems, incentive plans, factory layout problems, material handling problems, and principles of organization. Although these pioneers are too numerous to mention in this brief chronology, more complete historical accounts are available elsewhere.^{11–14}

The significance of the early development of industrial engineering is far greater than most people realize, and in fact, ranks among the greatest achievements of all time. In 1968, Peter Drucker¹⁵ described the importance of "scientific management," which we today would call industrial engineering:

Scientific management (we today would probably call it "systematic work study," and eliminate thereby a good many misunderstandings the term has caused) has proved

¹¹Ross W. Hammond, in H. B. Maynard (ed.), *Industrial Engineering Handbook*, 3rd ed. (New York: McGraw-Hill Book Company, 1971), pp. 1–3 through 1–17.

¹²John A. Ritchey (ed.), *Classics in Industrial Engineering* (Delphi, Ind.: Prairie Publishing Company, 1964).

¹³W. R. Spriegel, and C. E. Myers (eds.), *The Writings of the Gilbreths* (Homewood, Ill.: Richard D. Irwin, Inc., 1953).

¹⁴L. Urwick (ed.), *The Golden Book of Management* (London: Newman Neame Ltd., 1956).

¹⁵Peter Drucker, *The Age of Discontinuity* (New York: Harper & Row, Publishers, 1968), p. 271.

to be the most effective idea of this century. It is the only basic American idea that has had worldwide acceptance and impact. Whenever it has been applied, it has raised the productivity and with it the earnings of the manual worker, and especially of the laborer, while greatly reducing his physical efforts and his hours of work. It has probably multiplied the laborer's productivity by a factor of one hundred.

Shown in Figure 1.2 are a number of significant events and developments that have occurred in the evolution of industrial engineering. The position of each event relative to the time axis is intended only to show the approximate time at which that event occurred. In many cases, such as *time studies*, the event on the chart merely indicates its beginning; time studies are still a fundamental tool of industrial engineering.

At the top of the chart are shown four overlapping periods of emphasis. The labels are somewhat arbitrary. The period from approximately 1900 through the mid-1930s is generally referred to as *scientific management*. The next period, labeled *industrial engineering*, begins in the late 1920s and is shown extending to the present time. The period marked *operations research* had a great influence on industrial engineering practice and is shown beginning in the mid-1940s and extending somewhat past the mid-1970s. The fourth period, *industrial and systems engineering*, is shown beginning around 1970 and extending indefinitely into the future.

Notice that each period blends into the succeeding period. In this context, a particular period never really "ends." Instead, the cumulative influences of successive periods simply result in the attainment of the particular characteristics of following periods. The influences of Taylor and other pioneers certainly continue to the present time and are reflected in industrial engineering practices.

The authors believe that the profession has evolved to the point that a significant new direction has been emerging for several years. We envision this new orientation as not only building on the previous foundations of industrial engineering and operations research, but also including appropriate concepts from feedback control theory, computer science, behavioral theory, *systems engineering*, and cybernetics.

Some of the most exciting and productive years for our profession are just ahead. The opportunities for service to humankind have never been greater.

1.10. Industrial Engineering Organizations

Much can be learned about any profession by tracing the organizations which members of the profession form and/or join. The American Society of Mechanical Engineers provided the first forum for a discussion of the works of the early pioneers, particularly Taylor and his associates. Then, in 1912 the Society to Promote the Science of Management was formed. The name was changed in 1916 to the Taylor Society.

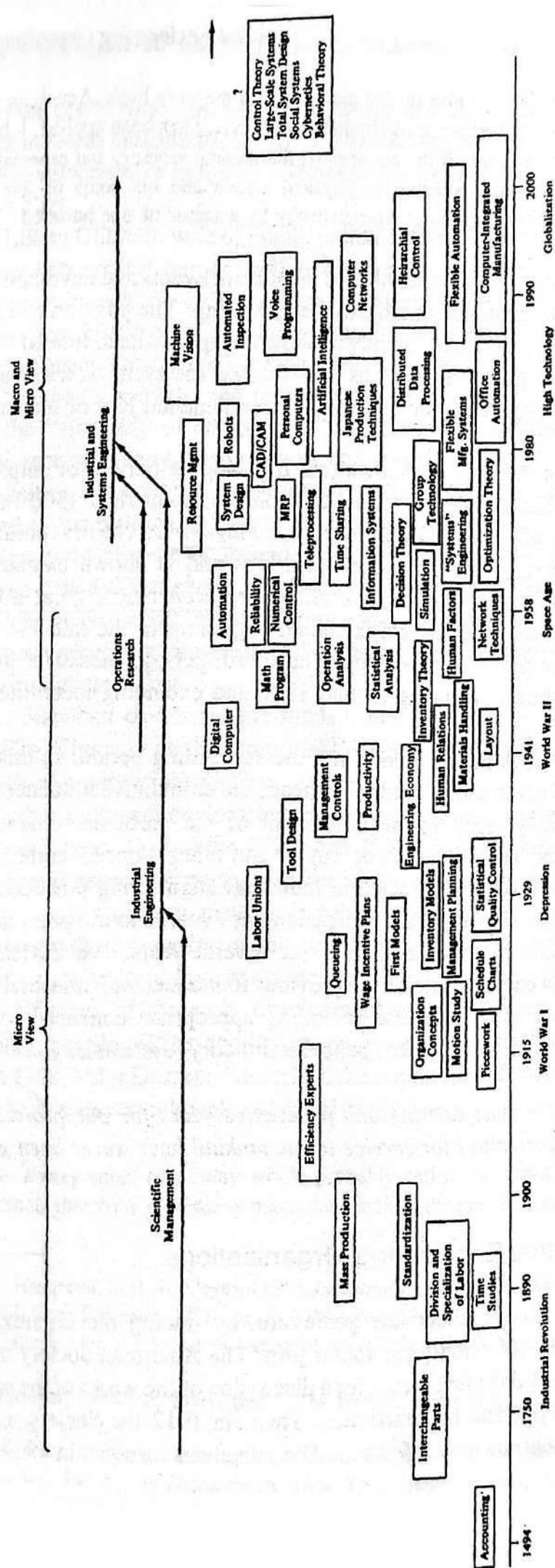


Fig 1.2. Chronology of significant events and developments in the evolution of industrial and systems engineering.

The Society of Industrial Engineers was formed prior to 1920. The American Management Association was formed in 1922, and many industrial engineers were active in this organization. In 1936 the Taylor Society and the Society of Industrial Engineers were combined to form the Society for the Advancement of Management.

The American Institute of Industrial Engineers was founded in 1948. The AIIE provided, for the first time, a professional organization devoted exclusively to the interests and development of the industrial engineering profession. Previously, I.E.s were associated with organizations whose main interests were in management or another branch of engineering.

The AIIE was an instant success in all respects. Within one year student chapters were formed at 11 major universities. The official publication of the AIIE, *The Journal of Industrial Engineering*, also made its initial appearance the following year, the first issue being published in June, 1949. In 1969 the *Journal* was divided into two publications. *Industrial Engineering* is published monthly and is devoted primarily to industrial engineering practice. *IIE Transactions* is published quarterly and is devoted primarily to research and new developments within the profession.

In 1981, through a vote of its membership, the Institute changed its corporate name from AIIE to IIE. By dropping the word "American," the Institute officially recognized the international nature of its activities. The IIE has members in more than 80 countries around the world.

Many practicing industrial engineers belong to other organizations that are related to the I.E. field. Some of these are:

- Operations Research Society of America.
- The Institute for Management Sciences.
- Association for Computing Machinery.
- American Society for Quality Control.
- Society for Decision Sciences.
- American Production and Inventory Control Society.
- Society of American Value Engineers.
- American Association of Cost Engineers.
- Society of Manufacturing Engineers.
- Robot Society of America.

It is significant that in 1970 the AIIE conducted a vote of its membership on a proposal to change the name of the institute to Institute of Industrial and Systems Engineers. Although this proposal was narrowly defeated (46% for; 54% opposed),¹⁶ it indicates that a sizable portion of IIE membership view their professional activities as concerned with broad aspects of total system design. This concept will be developed further in Chapter 2.

The IIE is the technical society for all industrial engineers, beginning with university students who are majoring in industrial engineering. A very important part of the overall organization are the university chapters. Students participating

¹⁶Frank Cotton, "On the Institute Name," *Industrial Engineering*, 2 (1970), 49.

in these chapters receive the IIE publications and are considered an integral part of the overall organization.

Several nationwide contests among students are conducted each year, such as the Outstanding Technical Paper Award. The winners of the several contests are recognized and honored at the annual conference of the IIE.

By beginning participation in his or her technical society as a student, a person is more likely to continue participation in professional activities after graduation.

* 1.11. Definition of Industrial Engineering

The following formal definition of industrial engineering has been adopted by the IIE:

* Industrial Engineering is concerned with the design, improvement, and installation of integrated systems of people, materials, information, equipment, and energy. It draws upon specialized knowledge and skill in the mathematical, physical, and social sciences together with the principles and methods of engineering analysis and design to specify, predict, and evaluate the results to be obtained from such systems.

As used in this context, the term *industrial* is intended to be interpreted in the most general way. Although the term *industrial* is often associated with manufacturing organizations, here it is intended to apply to any organization. The basic principles of industrial engineering are being applied widely in agriculture, hospitals, banks, government organizations, and so forth.

Read the definition again. Think about any large factory that you have seen in which thousands of workers, hundreds of machines, a large variety of materials, and millions of dollars must be combined in the most productive, cost-effective manner. Think about a large city that also requires thousands of workers, hundreds of vehicles and other machinery, materials, and millions of dollars in order to deliver services required by the public. Imagine how much more effectively the city could be run if the principles of industrial engineering were applied.

Consider again the right-hand side of Figure 1.2. The current definition of industrial engineering needs little or no modification to be a suitable definition for "industrial and systems engineering" in which the overall system design is emphasized.

1.12. Industrial Engineering Education

Topics that later evolved into industrial engineering subjects were initially taught as special courses in mechanical engineering departments. The first separate departments of industrial engineering were established at Pennsylvania State University

and at Syracuse University in 1908.¹⁷ (The program at Syracuse was short-lived, but it was reestablished in 1925.) An I.E. option in mechanical engineering was established at Purdue University in 1911. A rather complete history of industrial engineering academic programs may be found in Chapter 5 of *Origins of Industrial Engineering*.¹⁸

The practice of having an I.E. option within a mechanical engineering department was the predominant pattern until the end of World War II. Following World War II, separate I.E. departments were established in colleges and universities throughout the country.

There was very little graduate level work in industrial engineering prior to World War II. Once separate departments were established, master's level programs began to appear.

By 1990 over 150 universities were offering curricula in industrial engineering, of which 92 were accredited by the Accreditation Board for Engineering and Technology.¹⁹ A large number of the accredited programs also offer the master's degree.

Graduate programs in industrial engineering leading to the doctoral degree are recent developments. Prior to 1960 fewer than 100 total doctoral degrees in industrial engineering had been granted. By 1990 approximately 175 students received the doctoral degree each year.²⁰ Many doctoral degree holders go into teaching, but more and more are going into positions in industry and government.

1.13. Impact of Related Developments

The evolution of the industrial and systems engineering profession has been affected significantly by a number of related developments. We discuss several of these in this section.

1.13.1 Impact of Operations Research

The development of industrial engineering has been greatly influenced by the impact of an analysis approach called *operations research*. This approach originated in England and the United States during World War II and was aimed at solving difficult war-related problems through the use of science, mathematics, behavioral science, probability theory, and statistics. The approach enjoyed rather good success.

Following World War II the concepts of operations research were extended

¹⁷Hammond, pp. 1-15.

¹⁸Emerson and Naehring, pp. 43-58.

¹⁹1990 Annual Report, Accreditation Board for Engineering and Technology, New York, Sept. 1990, pp. 69-70.

²⁰Records of the Council of Industrial Engineering Academic Department Heads.

to problems in industry and commerce. A large number of mathematicians and scientists began devoting attention to a wide variety of operational problems. This resulted in considerable interaction between industrial engineers and members of other scientific disciplines. The infusion of new ideas and new approaches to problem solving had a dramatic impact on industrial engineering education and practice.

Industrial engineering departments at many universities began offering course work and complete options (usually at the graduate level) in operations research. In fact, some academic I.E. departments have changed their names to Industrial Engineering and Operations Research.

As it was originally conceived, the operations research approach was as follows: A specific problem was identified; specialists from appropriate fields were formed into an interdisciplinary task force to develop a solution; appropriate scientific methods and principles (usually involving mathematical models) were brought to bear on the problem; consideration was given to the interaction of the various components in the system being studied; the "best" solution was decided upon and presented to management. Once the work of the task force was completed, the task force was disbanded.

An approach such as this essentially precludes the establishment of operations research as a separate discipline. One person simply cannot have expertise in all the scientific areas that might be needed for problem solving.

As currently used, the term operations research connotes a set of quantitative methods that are applicable to a wide range of managerial and operational problems. Consequently, very few people consider themselves as operations researchers. Many industrial engineers do consider operations research as being their primary interest.

1.13.2 Impact of Digital Computers

Another development that has had a significant impact on the I.E. profession is the digital computer. Digital computers permit the rapid and accurate handling of vast quantities of data, thereby permitting the I.E. to design systems for effectively managing and controlling large, complex operations.

Many of the methods of operations research, discussed in the preceding section, require extensive calculations. The digital computer permits the I.E. to exercise a wide variety of optimization techniques to assist decision makers to better allocate scarce resources.

The digital computer also permits the I.E. to construct computer simulation models of manufacturing facilities and the like in order to evaluate the effectiveness of alternative facility configurations, different management policies, and other management considerations. Computer simulation is emerging as the most widely used I.E. technique.

A recent development that is having a profound (although still uncertain) impact on industrial engineering is computer-aided design (CAD) and computer-aided manufacturing (CAM). The I.E. is now designing facilities and work spaces directly on a computer workstation. The computer is also being used to automatically

generate process plans, bills of material, tool release orders, work schedules, operator instructions, and so on. During production, computers are used to control the cutting path of machine tools, send instructions to robots and other devices, record production progress, dynamically reschedule work centers, and automatically generate reports to all levels of management.

The development and widespread utilization of personal computers is having a dramatic impact on the practice of industrial engineering. I.E.s use PCs extensively to perform various analyses, to execute mathematical models, to plan and manage complex projects, to create data bases, and to implement a wide range of decision support tools.

Computer networks are being developed for companies that will permit all computers (even those made by different companies) to "talk to each other." In this mode, managers, supervisors, and machine operators may gain access to vast amounts of information that is stored in different computer systems, but accessible through the network.

The I.E.s role in the computer environment described above is to perform the overall system design, which effectively links together the various components of the system. Essentially every I.E. tool and technique discussed in the remainder of this book will be affected by developments in computers and communications.

1.13.3 Emergence of Service Industries

In the early days of the industrial engineering profession, I.E. practice was applied almost exclusively in manufacturing organizations. After World War II there was a growing awareness that the principles and techniques of I.E. were also applicable in nonmanufacturing environments.

One of the first service industries to utilize industrial engineering on a broad scale was the health-care industry. Many hospitals and clinics employ I.E.s to improve their operations, eliminate waste, control inventories, schedule activities, and for a wide variety of other functions.

A more recent area to utilize I.E.s is that of government agencies. This is now occurring at the national, state, and local levels. Thousands of I.E.s are employed by government organizations to increase efficiency, reduce paperwork, design computerized management control systems, implement project management techniques, monitor the quality and reliability of vendor-supplied purchases, and for many other functions.

Many other service industries are utilizing industrial engineers to improve their operations. The demand for I.E.s in all types of industry is far greater than the supply.

1.14. Relationship to Other Engineering Disciplines

As discussed in Section 1.3, the industrial engineering profession evolved from other engineering disciplines. Figure 1.3 illustrates the general nature of this evolution as well as the inputs to industrial engineering from certain nonengineering disciplines.

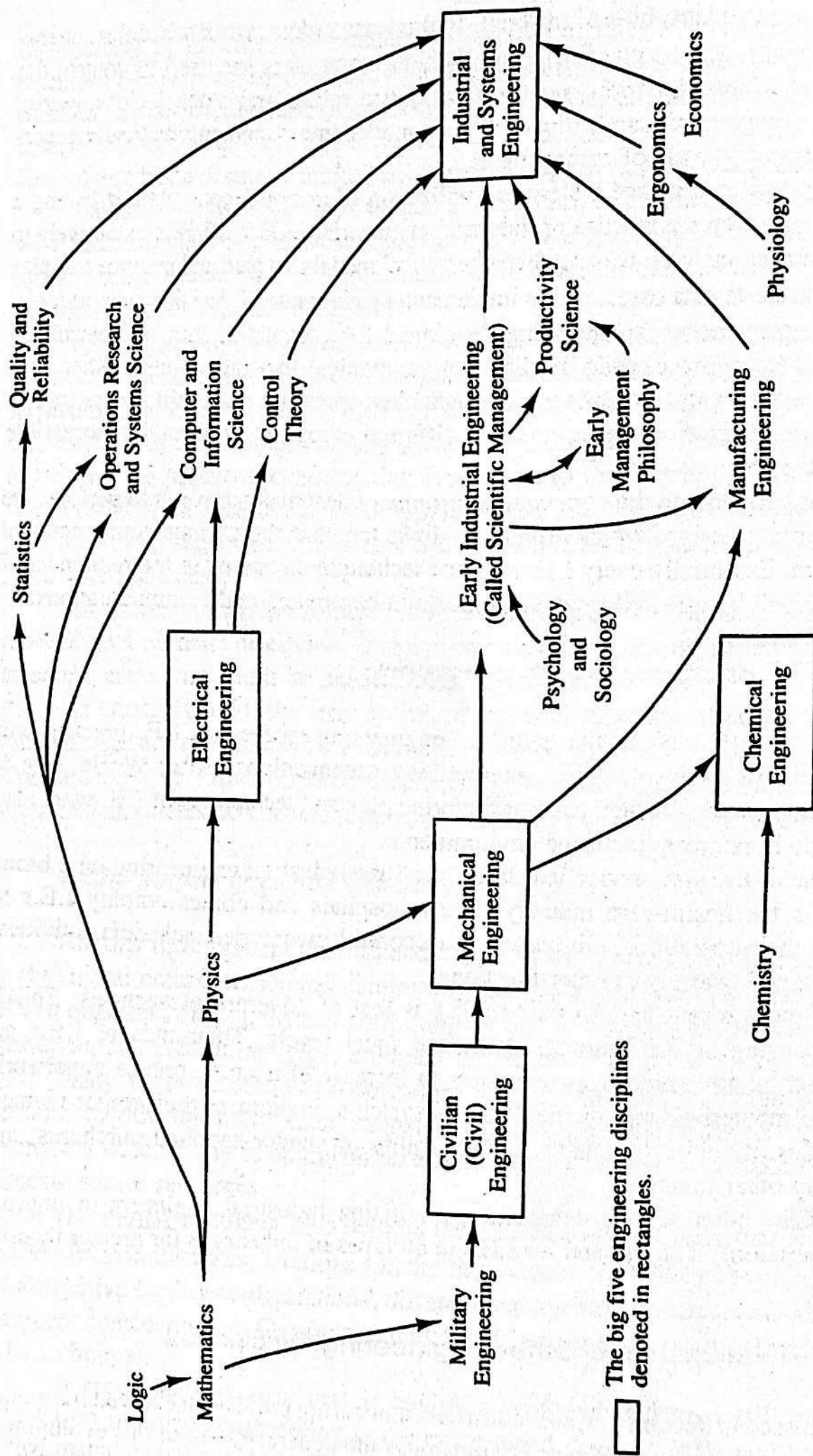


Fig. 1.3. Relationship of industrial and systems engineering to other engineering and scientific disciplines.

1.15. Challenges of the Future

We have briefly reviewed the role of the engineer in the technological developments that have led to the complex world in which we live. The remarkable achievements of the engineering profession have, ironically, contributed to some of the problems now facing human society. In the petroleum industry, engineers have designed highly efficient processes and systems for locating, extracting, processing, and distributing petroleum products. Increased efficiency led to increased usage. Increased usage led to increased profits, which led to even greater efficiency. This process could continue indefinitely if there were unlimited amounts of oil to be found. The oil shortages of the 1970s have made the world painfully aware that petroleum reserves are indeed limited.

Similar examples could be cited for other natural resources. The world has a finite amount of nonrenewable resources. One of the major challenges for future engineers is to learn to accomplish the engineer's mission in recognition of these constraints.

Another major challenge facing the engineering profession is to design systems and processes that are compatible with our natural environment. Dumping wastes into a river or allowing harmful gases to escape into the atmosphere are no longer permissible design strategies.

A major challenge that future engineers will encounter is that of designing products that are safe and reliable. More and more companies are being held accountable for faulty products that result in injury or harm to the buyer or to the public. Some states are now enacting legislation requiring that the design of all products affecting the public be approved by a registered professional engineer.

Through the application of sound industrial engineering practices, American industry became the strongest industry in the world during the two decades following World War II. Through the application of technological advancements and the never-ending search for improved work methods, American industry attained the highest level of productivity in the world. American products were regarded in the world market as being of the highest quality and reliability.

Productivity is the ratio of outputs to inputs. If factory A produces a 19-inch television set using 13.6 labor hours, and factory B produces an equivalent 19-inch television set using only 6.8 labor hours, then factory B has attained twice the productivity level of that attained by factory A. This concept will be developed further in Chapter 6.

American industry continues to lead the world in labor productivity, but other countries are rapidly closing the gap. Japan, France, West Germany, and other European countries are far outpacing the United States in the annual rate of productivity increase. Since labor costs in many of these countries are still below those in the United States, products manufactured in these countries can be sold on the world market at prices considerably below those of U.S. products. Consequently, the United States has lost large shares of the world market in many industries. This has had a negative impact on our balance of payments, inflation, and national debt.

Another major problem facing U.S. industry is the declining quality of American products relative to those made in many other countries. This has resulted, in part, from the aging equipment and processes in our factories. American industry, for a variety of reasons, has not made adequate investments in new technology, new equipment, and new manufacturing concepts.

There are encouraging signs in the early 1990s that American industry is beginning to take the steps necessary to regain a competitive position in world markets. An increasing number of companies are investing in automation, robots, and computer-controlled manufacturing processes. There is also an increasing emphasis on improved quality. The critical importance of a well-educated work force is also being recognized.

The business magazines are now saying that a “second industrial revolution” is under way in American industry. The engineering profession, particularly industrial engineering, will have a major role to play in this movement. We will look in greater detail at some of the aspects of this movement in Chapter 10.

As legislative leaders grapple with the many problems facing humankind, a realization will soon emerge that their solutions require an input from technically competent professionals. For Congress to enact a technologically infeasible national energy policy would be a disservice to the nation. *Perhaps the greatest challenge facing the engineering profession is to become involved in political issues and to provide assistance to lawmakers in the design of social systems.*

Despite the many remarkable achievements of engineers in the past, their greatest contributions to society will occur in the future. They will be expected to design systems that optimally utilize available resources for the satisfaction of human needs. Thus, the basic mission of the engineer remains essentially the same as always. Additional constraints, however, such as resource limitations, environmental concerns, and political issues, must be considered as an integral part of the engineering process.